# Deck Build Guide

Percolation Analysis of Vein Reticulation as a Structural Basis for Damage Tolerance in Leaf Venation Networks under Simulated Vascular Damage

Nicko Fajardo | 16 core slides + 7 hidden appendix | companion to DEFENSE_SCRIPT.pdf

Slide numbers below map to the numbered blocks of `DEFENSE_SCRIPT.pdf`. Every figure path was verified against the working tree when this PDF was generated. Every number was taken from Part 8 of `STUDY_MASTERY_GUIDE.md`.

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# 1. Deck specification

| Setting | Value |
|---|---|
| Aspect ratio | 16:9, 1920 x 1080 |
| Slide count | 16 core, 7 appendix (hidden) |
| Spoken runtime | 14 min 45 s across 10 script blocks |
| Longest blocks | 8 (2:35), 10 (2:05), 9 (1:45) |
| Margins | 80 px on all four sides |
| Export | PPTX and PDF; PDF is the projector fallback |
| Fonts | embed: File, Options, Save, Embed all characters |

### Type

| Role | Font | Size |
|---|---|---|
| Slide title | Inter SemiBold (fallback Calibri) | 40 pt |
| Body | Inter Regular | 24 to 28 pt, never below 22 |
| Hero number | Inter Bold | 90 to 140 pt, one per slide |
| Hero caption | Inter Regular | 20 pt |
| Figure axis text | from matplotlib | 18 pt equivalent minimum |

### Colour

| Token | Hex | Applied to |
|---|---|---|
| Ink | #1A1D21 | body text, titles |
| Paper | #FAFAF7 | slide background |
| Vein | #2E7D5B | connectivity, reticulation, the intervention |
| Loss | #C1462F | damage, removal, collapse, thresholds |
| Muted | #6B7280 | captions, secondary labels, sources |
| Rule | #E2E2DC | hairlines, table borders |

Green means connectivity and red means damage on every slide, without exception. Pair each colour with a label or position as well, since roughly one panelist in twelve cannot separate the two by hue.

### Species naming

The study models a generic haustorial leak conductance, not a species. The manuscript names *Cuscuta* in three roles only: introduction motivation, the pre-symptomatic time-course literature, and the scope of the Balios et al. 2025 published negative. The calibrated 18 to 20 percent diversion is a hemiparasite measurement (Jiang et al. 2003).

A species name therefore appears on **slides 2, 16 and A1 only**, and on no methods, results or model slide.

## 2. Slide-by-slide

### SLIDE 1 Title before block 1

`figures/board_photos/cuscuta_b.jpg`

**Layout** full-bleed photo darkened to about 35 percent, text overlaid bottom-left

**Content** full study title at 32 pt over two lines; name, school, adviser at 20 pt Muted

**Exclude** outline, agenda, objectives

### SLIDE 2 Opening image block 1

`figures/board_photos/haustorium.jpg`

**Layout** full-bleed image, one text line bottom-left at 28 pt

**Line** *Cuscuta*. A holoparasite. It drills.

**Constraint** the word holoparasite must be present; this is one of three slides permitted a species name

### SLIDE 3 The negative result block 2

**Layout** hero number, centred

**Hero** 2.6% in Loss red at 140 pt

**Caption** the modelled sink's share of the safety margin (not "the parasite's")

**Strip** 0.026 MPa depression | 6.0 MPa needed to cavitate | 0.5 MPa largest ever measured

**Build** strip appears on click

### SLIDE 4 Branching against reticulate block 3

`figures/methodology/fig_reticulation_concept.png`

**Layout** two panels, 50/50

**Left label** Branching, one route (Loss red)

**Right label** Reticulate, detour available (Vein green)

**Footer** Sack et al., midrib cut = -75% conductance; minor veins cut = no detectable penalty

**Build** left panel, then right on click

**Blocking** the script points left for branching and right for reticulate; keep those sides on slide 14 as well

### SLIDE 5 Fragmentation block 4

`figures/collapse_montage.png`

**Layout** single figure, full width

**Title** Connectivity does not decay. It shatters.

**Preferred** figures/collapse.gif, a 6-second loop on a real cohort leaf; confirm playback on the venue machine

**Fallback** figures/collapse_montage.png if the venue machine will not loop a GIF

**Regenerate** scripts/make_collapse_gif.py

### SLIDE 6 Definitions block 4

**Layout** three columns

**Columns** Nodes = vein junctions | Edges = vein segments | Threshold = fraction removed before the giant component halves

**Footer** operational, not a physical critical point; source-anchored cross-check rho = 0.912

**Range to state if asked** 421 to 300,633 nodes

## SLIDE 7 Data and its limitation block 5

`figures/methodology/fig02_dataset.png`

**Layout** figure left 60 percent, facts right 40 percent

**Facts** 120 traced networks, 122 species, Matos et al. 2024, Dryad

**Cohorts** 112 characterisation, 100 percolation, 105 enhancement

**Boxed in Loss red** 100 of 120 are calibrated crop windows, not whole leaves

## SLIDE 8 Methods pipeline block 5

`figures/methodology/fig01_pipeline_revised.png`

**Layout** full-width horizontal flow, title only, no bullets

**Note** the one slide permitted to be dense; regenerate if internal text falls below 18 pt on screen

## SLIDE 9 The threshold block 6

`figures/sigmoid.png`

**Layout** hero number left 40 percent, figure right 60 percent

**Hero** 0.1998 at 120 pt in Loss red

**Caption** median critical removed fraction, n = 100

**Footer** 2,400 Monte Carlo trials per network per mode, fixed seeds

**Build** none

## SLIDE 10 Degree-weighted removal block 6

`figures/sigmoid.png`

`figures/sigmoid_targeted.png`

**Layout** two collapse curves side by side on shared axes

**Numbers** 0.1998 to 0.1685, -15.7 percent, 100 of 100 networks

**Boxed footer** not an attack; probabilistic weighting, a bounding scenario

## SLIDE 11 Architecture predicts tolerance block 7

`figures/cohort_pc_predictors.png`

**Layout** figure top 65 percent, quartile table bottom 35 percent

**Correlations** mean degree rho = 0.733; fractal dimension rho = 0.496

**Size-controlled** 0.745 and 0.633, both stronger

**Model** R-squared = 0.507, F(2,97) = 49.93

**Build** figure, table on click, correlations on click

| Reticulation quartile | Loop fraction | Mean degree | Median threshold |
|---|---|---|---|
| Q1 least reticulate | 0.616 | 2.320 | 0.1584 |
| Q2 | 0.738 | 2.472 | 0.1850 |
| Q3 | 0.807 | 2.564 | 0.2087 |
| Q4 most reticulate | 0.901 | 2.734 | 0.2308 |

Q4 minus Q1 is 7.2 percentage points, a 45.7 percent relative gain, monotonic across all four quartiles. U = 4, delta = 0.987.

## SLIDE 12 The intervention block 8

`figures/cohort_suggested.png`

**Layout** figure centred, result overlaid bottom-right

**Result at 56 pt** 93 improved, 12 unchanged, 0 worse

**Numbers** 0.2010 to 0.2150, median difference 0.0095, p = 1.7e-18

**Method footer** Delaunay, parameter-free; capped at cohort P90 mean degree 2.811; common random numbers; 10,000 trials per arm

**Build** the "0 worse" appears last, on its own click

**Note** block 8 runs 2:35 and covers this slide and slide 13; do not rush off it

## SLIDE 13 The intervention, shown block 8

`figures/methodology/fig_realsuggested_Met_dav.png`

**Layout** before and after, side by side, identical scale

**Labels** Original | Plus loops, from its own coordinates

**Alternative if added edges are hard to see** figures/methodology/fig_realsuggested_Cis_ant_zoom_pair.png

## SLIDE 14 The scope limit block 9

`figures/locality_boxplots.png`

**Layout** boxplots left 60 percent, numbers right 40 percent

**Numbers** 0.1998 dispersed, 0.4951 one lesion (2.46x), 0.4511 ten lesions (2.27x); 100 of 100 both

**In Loss red** rho 0.737 dispersed, 0.151 ten lesions, -0.000 one lesion

**Footer** claim scoped to dispersed damage

**Build** the correlation collapse on click

## SLIDE 15 Scope as a claim block 9

**Layout** text only, centred, 44 pt, three lines. The only text-only slide in the deck.

**Line 1** Reticulation governs tolerance

**Line 2** to *dispersed* damage.

**Line 3** Under a single lesion, the effect is gone.

**Footer** fractal dimension behaves oppositely and survives multi-lesion damage at rho = 0.523

## SLIDE 16 Close block 10

`figures/board_photos/cuscuta_b.jpg`

**Layout** three lines, hairline, then two gap lines; small photo thumbnail bottom-right

**Line 1** about 20 percent of junctions to fragmentation

**Line 2** reticulation raises that threshold, causally, within-leaf

**Line 3** conditional on dispersed damage

**Gap line 1 at 24 pt** Cuscuta-host, no published water potential (Balios et al. 2025)

**Gap line 2 at 24 pt** holoparasites as a class, none found (this study's search)

**Constraint** the two gap lines stay separate and neither widens to "parasitic plants"; hemiparasite water relations have been measured, and the 20 percent on slide 3 is one of those measurements

**Exclude** a Thank you or Questions slide; the conclusion stays on screen through questioning

## 3. Builds and animation

Permitted: Appear with no duration, and Fade at 0.2 s or less. Not permitted: Fly In, Wipe, Zoom, Bounce, Morph, and slide transitions of any kind.

| Slide | Click | Reveals |
|---|---|---|
| 3 | 1 | the MPa strip |
| 4 | 1 | the reticulate panel |
| 5 | up to 3 | montage panels left to right |
| 11 | 2 | quartile table, then the correlations |
| 12 | 1 | 93 / 12 / 0 |
| 14 | 1 | the correlation collapse |

Total build clicks: 9 to 11 depending on slide 5. Every other slide appears whole. Write the per-slide click count on the notes card.

# 4. Figures

### Figures required, all verified present

| Slide | Path |
|---|---|
| 1, 16 | figures/board_photos/cuscuta_b.jpg |
| 2 | figures/board_photos/haustorium.jpg |
| 4 | figures/methodology/fig_reticulation_concept.png |
| 5 | figures/collapse.gif, figures/collapse_montage.png |
| 7 | figures/methodology/fig02_dataset.png |
| 8 | figures/methodology/fig01_pipeline_revised.png |
| 9, 10 | figures/sigmoid.png, figures/sigmoid_targeted.png |
| 11 | figures/cohort_pc_predictors.png |
| 12 | figures/cohort_suggested.png |
| 13 | figures/methodology/fig_realsuggested_Met_dav.png |
| 14 | figures/locality_boxplots.png |
| A2 | figures/methodology/fig12_stats_tree.png |
| A3 | figures/calibration_grid.png |
| A4 | figures/sink_demo.png, figures/sink_surface.png |
| A6 | figures/finite_size_fixedD.png |

### The collapse animation

`figures/collapse.gif` is generated by `scripts/make_collapse_gif.py`. It runs on a real cohort leaf rather than the synthetic mesh, and it draws the giant connected component in Vein green with everything detached from it in grey. That colour split is the whole point: with one colour for all surviving veins, the eye reads progressive thinning rather than fragmentation. Removal is nested from a single seeded permutation, so the sequence is one sweep rather than unrelated draws. Roughly 6 seconds, 1.1 MB, holds on the intact network at the start and the shattered state at the end.

The threshold line is that leaf's archived value. The animation is a single realisation, so its own crossing sits slightly later; the archived number is a median over 2,400 trials. Say "one run" if asked, not "the threshold".

### The locality figure

`figures/locality_boxplots.png` is generated by `scripts/make_locality_figure.py` from `data/criterion_and_locality.csv`, 100 rows, columns `pc_gcc` (dispersed), `pc_disc1` (one lesion) and `pc_disc10` (ten lesions). Re-run that script if the underlying data changes; it prints the medians and ratios it drew.

Each network is also drawn as a faint line carried across the three conditions, because the claim is that every network moved the same way rather than that the distributions differ on average, and only the paired lines show that. The 2.46x and 2.27x printed on it are median per-network ratios, not ratios of the medians; the two definitions differ in the third decimal and the per-network one is what the manuscript reports.

### Preparing the rest

Check each figure full-screen from three metres. If the axis labels cannot be read, regenerate at larger type rather than scaling the image, which only enlarges blurred text. Set before regenerating: font.size 16, axes.labelsize 18, axes.titlesize 20, tick labelsize 15, legend.fontsize 15, savefig.dpi 200, bbox tight, facecolor #FAFAF7.

Remove figure-internal titles, which duplicate the slide title, along with unused gridlines and single-entry legends. Recolour series to #2E7D5B and #C1462F. Write slide copies to `figures/slides/` and never overwrite a manuscript figure. Export PNG at 200 dpi or higher; JPEG artefacts around thin vein lines read as tracing noise.

# 5. Appendix slides

Hidden from the main flow with right-click, Hide Slide. Reached during questioning by typing the slide number and Enter. Write the numbers on the notes card.

| Slide | Contents | Question it answers |
|---|---|---|
| A1 | sink scoping card, contents below | which parasite are you modelling |
| A2 | figures/methodology/fig12_stats_tree.png | why Spearman and not Pearson |
| A3 | figures/calibration_grid.png | how were connectivity and D separated |
| A4 | figures/sink_demo.png and figures/sink_surface.png | show the sink model |
| A5 | cavitation threshold sweep | is zero embolised robust |
| A6 | figures/finite_size_fixedD.png | is this a phase transition |
| A7 | limitations table, contents below | what are your limitations |

### A1, sink scoping card

- Modelled: a generic haustorial leak conductance, K_h = 0.0392, giving 20.0 percent withdrawal
- Anchor: Jiang et al. 2003, a hemiparasite; no holoparasite flux fraction has been measured
- Gradient: about 1.5 MPa, stipulated, declared as a delimitation
- Attachment in nature: Cuscuta to stems, petioles and pedicels; Orobanche to roots; the networks here are leaf laminae
- In Loss red: a haustorial node inside a lamina is a modelling device, declared as one

### A5, cavitation threshold sweep

| Cavitation threshold | Onset gradient |
|---|---|
| -1.00 MPa | 1.1 MPa |
| -2.00 MPa | 6.0 MPa |
| -3.00 MPa | 9.5 MPa |

Zero vessels embolised holds from 0 to 3.0 MPa but inverts at a -1.00 MPa threshold, so the result is specific to thresholds at or below about -1.10 MPa within the swept range.

### A7, limitations

- No phylogenetic correction; species are treated as independent when they are not
- No multiple-comparison correction; the analysis is exploratory over a pre-specified predictor set
- 100 of 120 files are crop windows; no petiole flag exists in the deposit
- No construction-cost, mechanical or photosynthetic term in the enhancement experiment
- R-squared 0.507; about half the variance in the threshold is unmeasured
- K_leaf validation rho = 0.187, p = 0.073, underpowered and xylem-only

- Zero vessels embolised is threshold-dependent and inverts at -1.00 MPa
- Betweenness-weighted removal exists on the synthetic reference network only
- Bootstrap resample capped at 50,000 trials, so intervals on longer runs are conservative

# 6. Build order and checks

### Order

- 1. Slides 12, 14, 15
- 2. Slides 3 and 2
- 3. Slides 9, 10, 11
- 4. Slides 4, 5, 6, 7, 8
- 5. Slides 1 and 16
- 6. Appendix A1 and A7 first, then A2 to A6
- 7. Remove one element from every slide
- 8. Rehearse against a timer, aloud and standing

### Checks before the defense

- Species names appear on slides 2, 16 and A1 only
- Slide 16 keeps its two gap lines separate; neither says parasitic plants
- Slide 3 caption reads the modelled sink, not the parasite
- figures/locality_boxplots.png placed on slide 14
- Exported to PPTX and PDF, on a USB stick and in cloud storage
- Fonts embedded, all characters
- Opened once on a machine other than the one it was built on
- Every figure legible from three metres; no slide over six lines
- Click counts and appendix slide numbers written on the notes card
- Presenter view tested; notes are three-word cues, not sentences
- Slide 5 GIF confirmed to loop on the venue machine, or replaced with figures/collapse_montage.png
- No slide says attack, hydraulic collapse, validated, or leaves where it should say leaf venation networks

## 7. Related documents

| Document | Contents |
|---|---|
| DEFENSE_SCRIPT.pdf | the spoken script; rebuild with scripts/make_defense_script_pdf.py |
| PRESENTATION_SCRIPT.md | the full 40-question bank |
| STUDY_MASTERY_GUIDE.md | concepts, Part 5 do-not-claim, Part 8 fast facts |
| DEFENSE_PANEL_PREP.md | the ten questions the adviser flagged |
| MANUSCRIPT_REVISED_SECTIONS.md | canonical figure numbering and captions |
| rrl/E_parasitic_plants_haustoria.md | evidence behind the species-naming rule |